

PUBLIC EXHIBIT 14

EXHIBIT 14

Oura Ring 4 Infringement Claim Chart - U.S. Patent No. 10,945,677

	U.S. Patent No. 10,945,677
Independent Claim 1	Oura Ring 4 and Corresponding Applications (collectively, “Oura Accused Product”)
1[pre] 1. A wearable electronic device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable electronic device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

[\(https://ouraring.com/blog/oura-ring-4/\)](https://ouraring.com/blog/oura-ring-4/)

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#).
(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,945,677
1[a] a motion sensor;	The Oura Accused Product includes a motion sensor. For example, the Domestic Industry Products include a motion sensor that measures motion, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,945,677	
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1[b] one or more optical sensors includes a light emitting unit and a light receiving unit; and	The Oura Accused Product includes one or more optical sensors includes a light emitting unit and a light receiving unit. For example, the Oura Accused Product includes optical sensors, such as a light sensor, which includes a light emitting unit, such as a red and/or infrared (IR) LED, and a light receiving unit, such as a light sensor or photodetector.

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	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

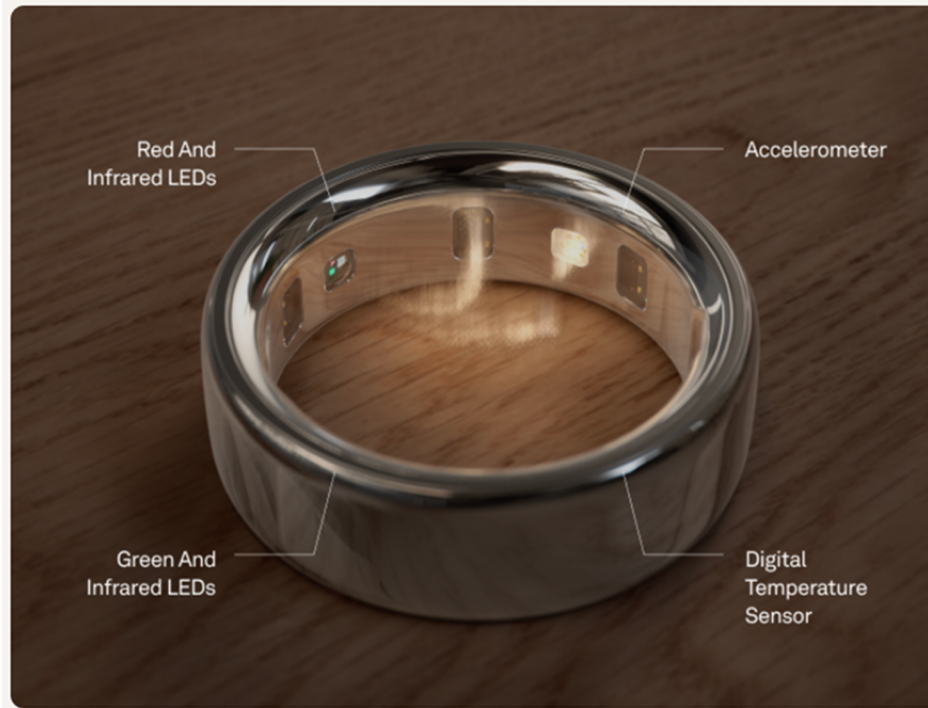
PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

<https://support.ouraring.com/hc/en-us/articles/4410651298963-Live-Heart-Rate>

Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

1[c] at least one processor configured to:

The Oura Accused Product includes at least one processor. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep.

This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

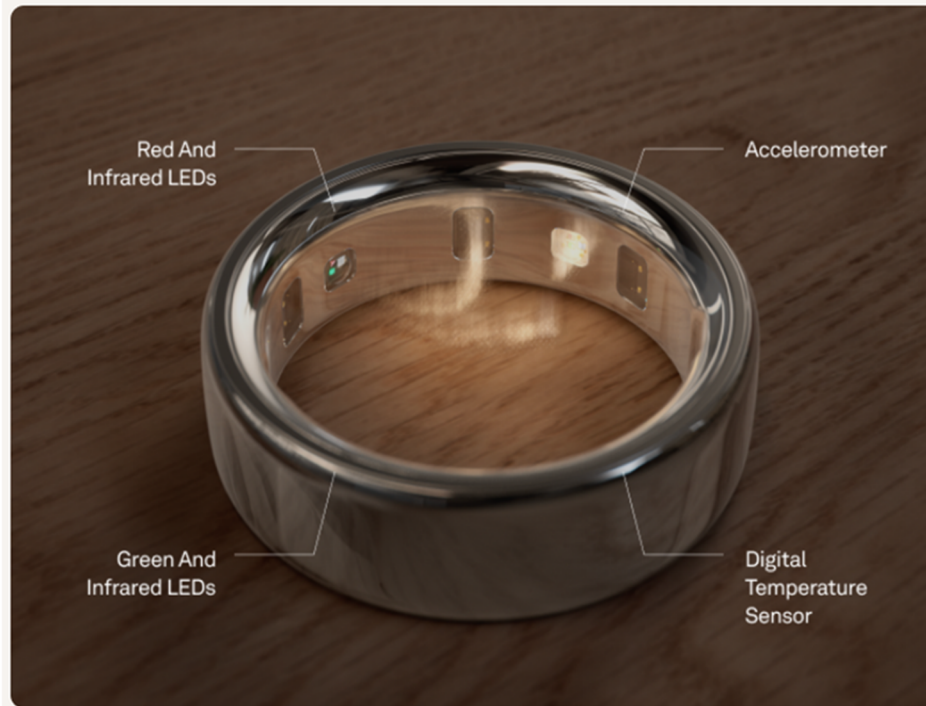


(<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>)

	U.S. Patent No. 10,945,677
1[c][i] obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,	The Oura Accused Product obtains, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors. For example the Oura Accused Product obtains, using the motion sensor that measures motion, such as an accelerometer, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors (<i>see</i> Element 1[c][ii]-[iii]). Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Ouraring-4)

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How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
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- Average body temperature drops

Deep sleep:

- Low movement
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REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

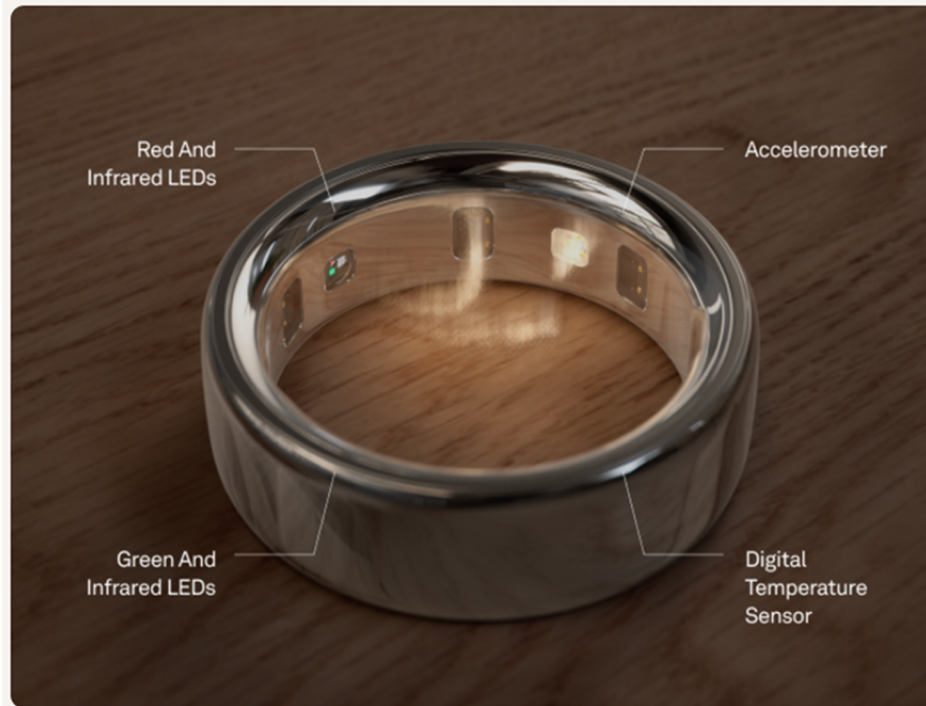
(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

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1[c][ii] emit light using the light emitting unit based at least in part on a determination	The Oura Accused Product emits light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition. For example, the Oura Accused Product emits light using the light emitting unit, such as using a light sensor, such as a red and/or infrared (IR) LEDs, based at least in

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that the first motion signal satisfies a specified condition,	part on a determination that the first motion signal satisfies a specified condition, such as the user being asleep or the motion signal satisfying a condition indicative of sleep. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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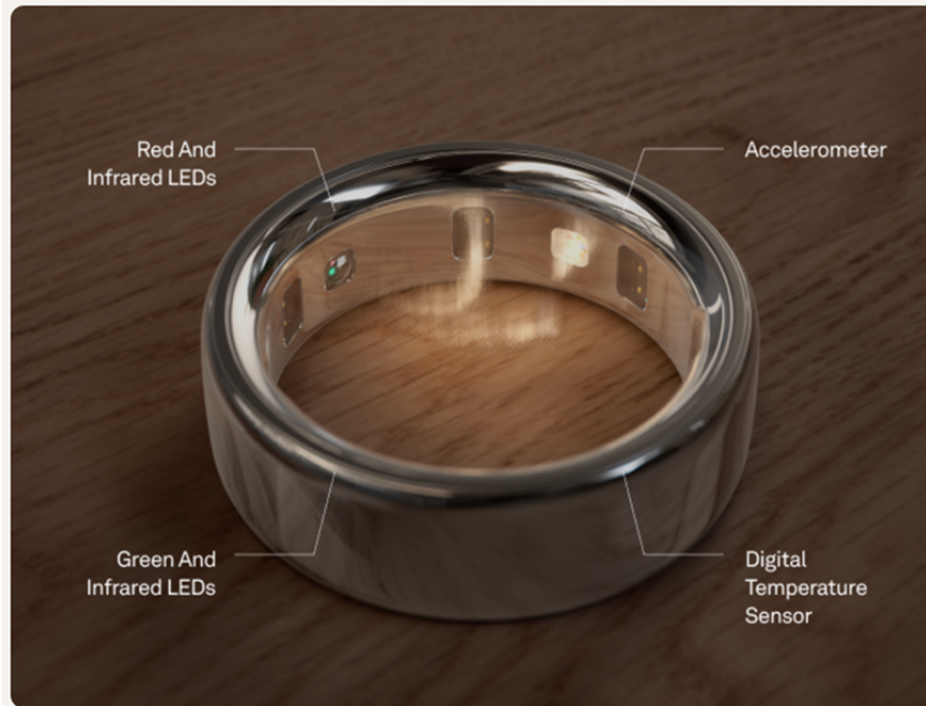
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1[c][iii] obtain the biological signal corresponding to a user using the light receiving unit based at least in part on	The Oura Accused Product obtains the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user. For example, the Oura Accused Product obtains the biological signal, such as blood oxygen (SpO ₂) levels, corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user.

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1[c][iv] obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device	The Oura Accused Product obtains, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal. For example, the Oura Accused Product obtains, using the motion sensor, a second motion signal corresponding to the motion of the wearable

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during the obtaining of the biological signal, and	electronic device, during the obtaining of the biological signal, such as blood oxygen (SpO₂) levels while, for example, the user is asleep. Sensors • Smart Sensing is a new sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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Learn more about Oura's sleep measurement on our blog: [How Does the Oura Ring Track My Sleep?](#)

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1[c][v] refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.	The Oura Accused Product refrains from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition. For example, the Oura Accused Product refrains from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition, such as a determination that the user is not asleep or the second motion signal satisfying a condition indicative of not being asleep.

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- Heart rate increases

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(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

	U.S. Patent No. 10,945,677
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2)
Independent Claim 9	
9[pre] 9. A wearable electronic device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable electronic device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

[\(https://ouraring.com/blog/oura-ring-4/\)](https://ouraring.com/blog/oura-ring-4/)

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ) and [Oura Ring Generation 3](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ).
(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

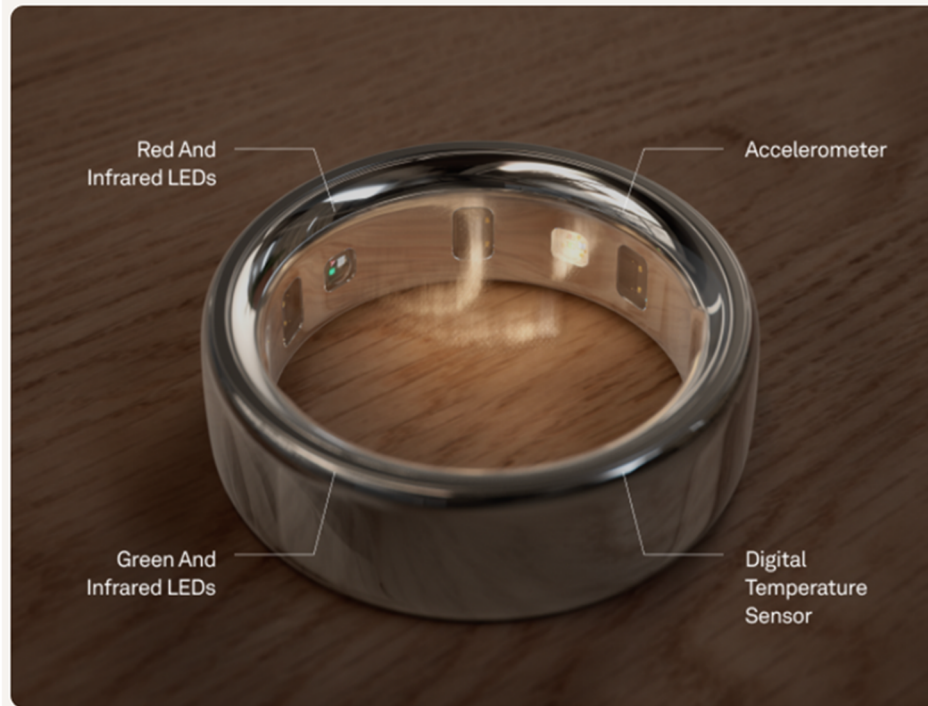
	U.S. Patent No. 10,945,677
9[a] a first sensor;	The Oura Accused Product includes a first sensor. For example, the Oura Accused Product includes a first sensor, such as a motion sensor and/or an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4)

U.S. Patent No. 10,945,677		
	Sensors • Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)	
9[b] a second sensor; and	The Oura Accused Product includes a second sensor. For example, the Oura Accused Product includes second sensors, for example optical sensors, such as the red and/or infrared (IR) LED and sensor that measure(s) biological information, such as blood oxygen (SpO₂) levels.	

U.S. Patent No. 10,945,677	
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,945,677			
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Oura Ring 4: More Accurate Than Ever Before



Crafted with precision, the [accuracy of Oura Ring](#) has been validated by research institutions across the globe, such as the University of Tokyo and the [National University of Singapore](#). In Oura Ring 4, the new Smart Sensing technology provides an unparalleled degree of accuracy and personalization.

This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

9[c] at least one processor configured to:

The Oura Accused Product includes at least one processor. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep.

This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.



(<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>)

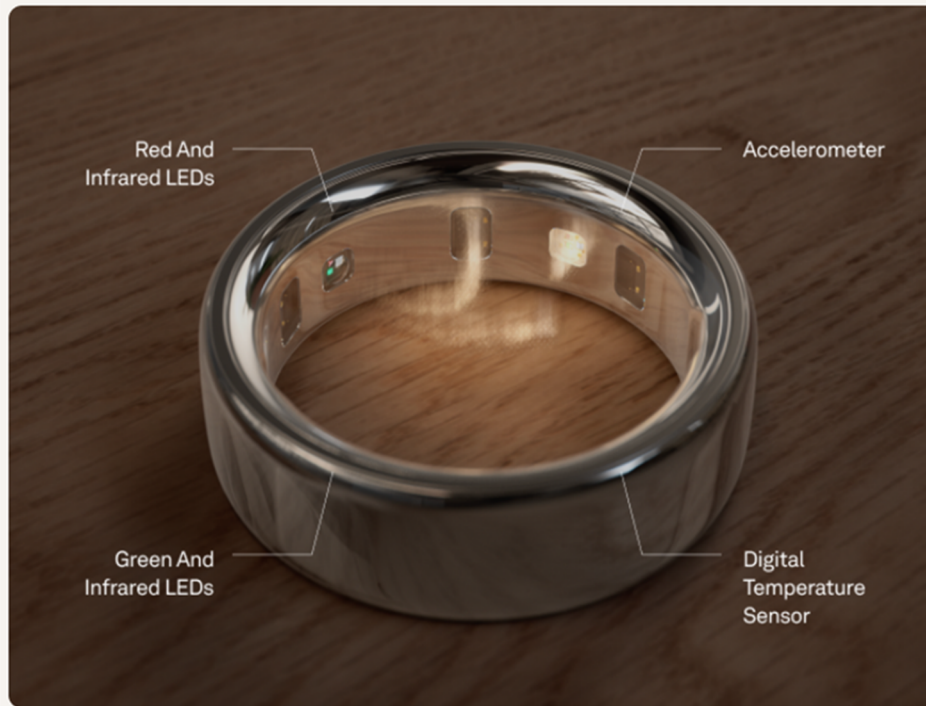
	U.S. Patent No. 10,945,677
9[c][i] obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,	The Oura Accused Product obtains, using the first sensor, first information corresponding to movement of the wearable electronic device. For example, the Oura Accused Product obtains, using the first sensor, a motion sensor such as an accelerometer, first information corresponding to movement of the wearable electronic device. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,945,677			
	<table border="1"> <tr> <td>Sensors</td><td> • Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7 </td></tr> </table> (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7
Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7		
9[c][ii] activate the second sensor based at least in part of the movement,	The Oura Accused Product activates the second sensor based at least in part of the movement. For example, the Oura Accused Product activates the second sensor, for example an optical sensor, such as a red and/or infrared (IR) LED and sensor, based at least in part of the movement, such as the user being asleep or the movement satisfying a condition indicative of sleep.		

U.S. Patent No. 10,945,677	
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,945,677			
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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

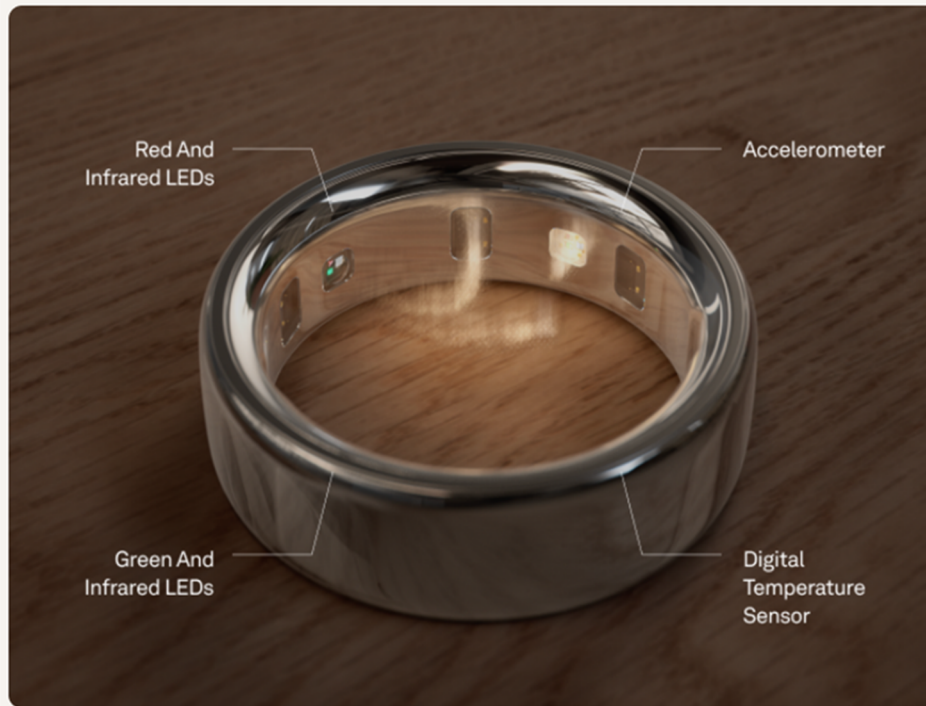
(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

	U.S. Patent No. 10,945,677
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9[c][iii] obtain, using the activated second sensor, second information corresponding to biological	The Oura Accused Product obtains, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information. For example, the Oura Accused Product obtains, using the activated second sensor, second information corresponding to a biological signal

	U.S. Patent No. 10,945,677
signal related to a user during the obtaining of the first information, and	related to a user, such as blood oxygen (SpO₂) levels, during the obtaining of the first information corresponding to movement of the device. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,945,677			
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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

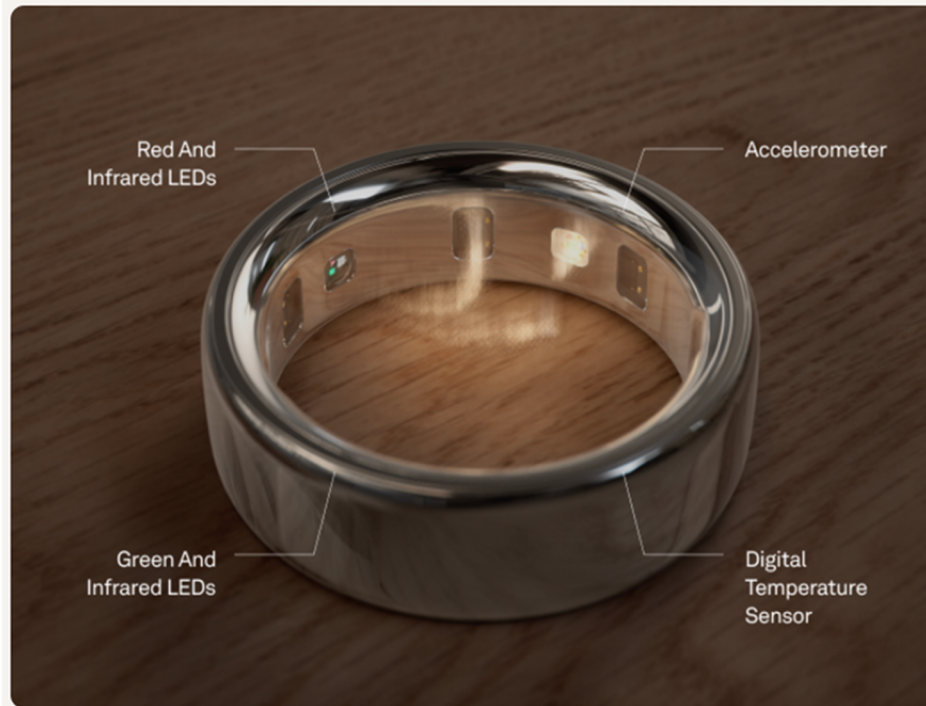
<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

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9[c][iv] refrain from the obtaining of the second information using the activated second sensor based at least in	The Oura Accused Product refrains from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range. For example, the Oura Accused Product refrains from the obtaining of the second information, for example biological information such as SpO ₂ or blood oxygen levels, using the activated second sensor based at least in

	U.S. Patent No. 10,945,677
part on a determination that the first information satisfies a specified range.	part on a determination that the first information satisfies a specified range, such as the user being awake or the user's motion satisfying a specified motion range indicative of being awake. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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Independent Claim 16	
16[pre] 16. A wearable electronic device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable electronic device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

[\(https://ouraring.com/blog/oura-ring-4/\)](https://ouraring.com/blog/oura-ring-4/)

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ) and [Oura Ring Generation 3](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ).
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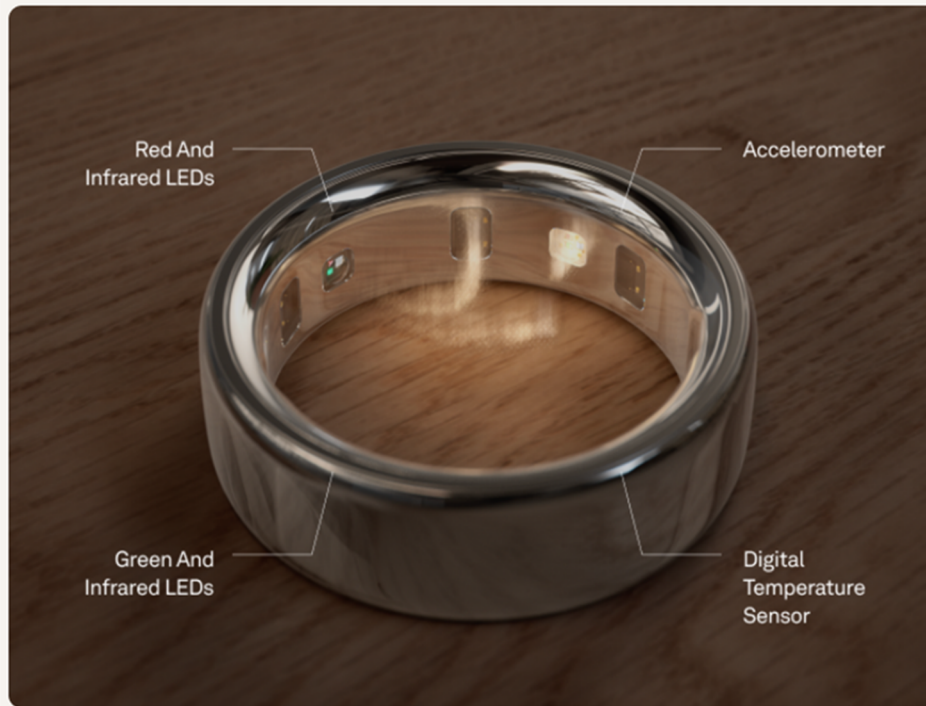
	U.S. Patent No. 10,945,677
16[a] a first sensor;	The Oura Accused Product includes a first sensor. For example, the Oura Accused Product includes a first sensor, such as a motion sensor and/or an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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	https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4		
16[b] a second sensor; and	The Oura Accused Product includes a second sensor. For example, the Oura Accused Product includes a second sensor, for example optical sensors, such as the red and/or infrared (IR) LED and sensor that measure(s) biological information, such as blood oxygen (SpO ₂) levels.		

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This breakthrough health sensing platform works with the ring's research-grade sensors to learn your unique physiology. The multiple sensors inside the ring form an 18-path multi-wavelength photoplethysmography (PPG) subsystem to provide highly accurate, continuous biometric data, including resting heart rate, [sleep stages](#), [heart rate variability \(HRV\)](#), temperature trends, and more, throughout the day and night.

<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

16[c] at least one processor configured to:

The Oura Accused Product includes at least one processor. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.



(<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>)

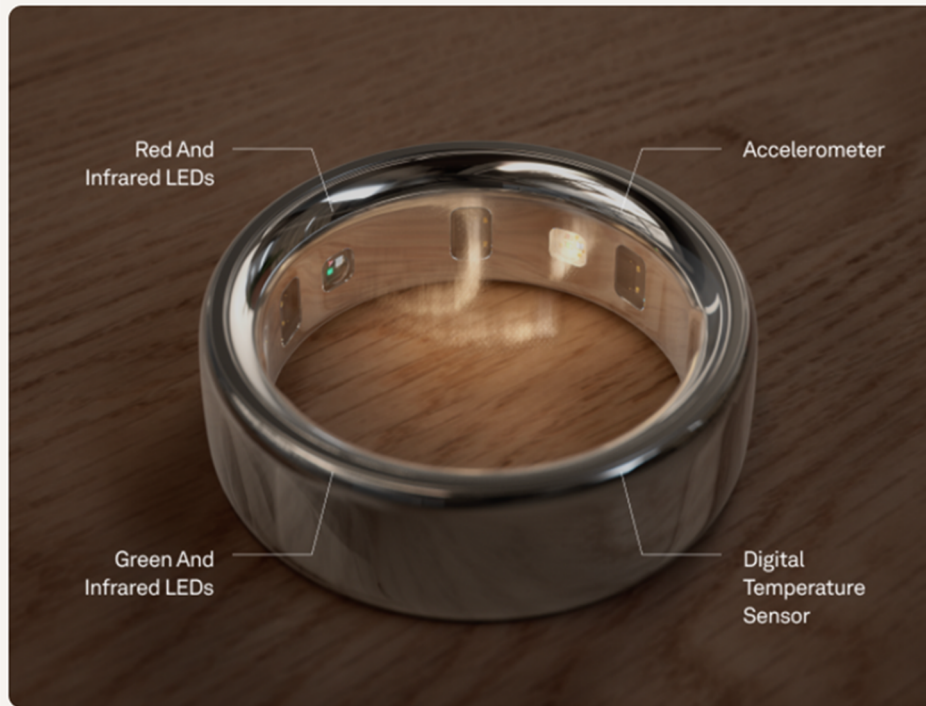
	U.S. Patent No. 10,945,677
16[c][i] obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,	The Oura Accused Product obtains, using the first sensor, first information corresponding to movement of the wearable electronic device. For example, the Oura Accused Product obtains, using the first sensor, a motion sensor such as an accelerometer, first information corresponding to movement of the wearable electronic device. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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16[c][ii] obtains, using the second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and;	The Oura Accused Product obtains, using the second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information. For example, the Oura Accused Product obtains, using the second sensor, for example optical sensors, such as the red and/or infrared (IR) LED and sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, such as blood oxygen (SpO ₂) levels.		

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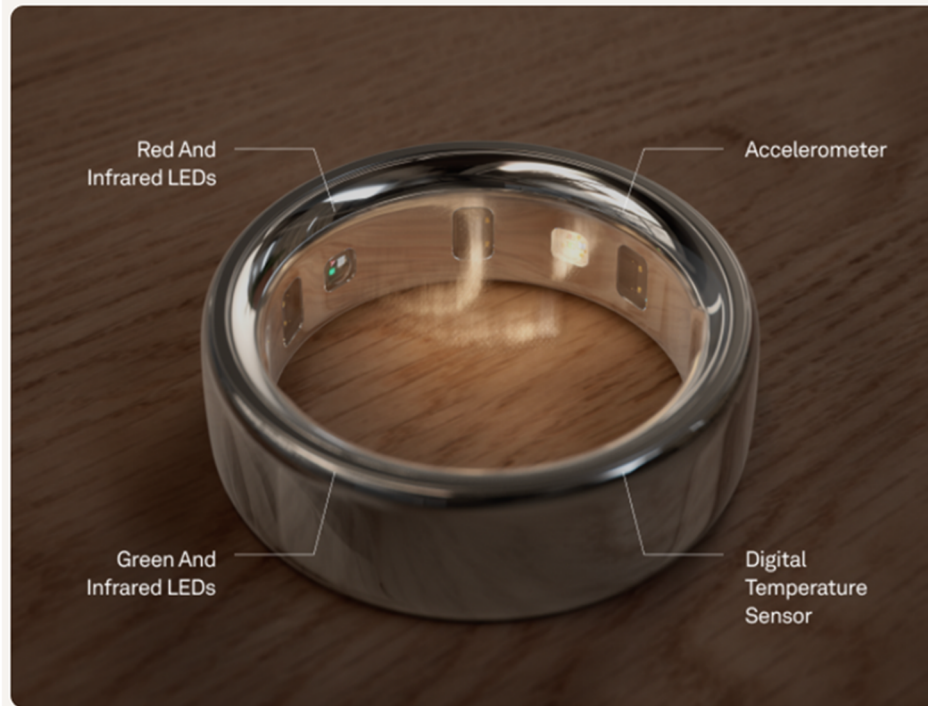
<https://ouraring.com/blog/oura-ring-4/?srsltid=AfmBOoprNrYcsfQDaXXRORngQyVU-anCvQA1xy643RvYSDEZ7fIDmo8Y>

	U.S. Patent No. 10,945,677
	How Does Blood Oxygen Sensing (SpO2) Work? The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDS into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light. Your blood oxygen measurements are located on the Oura App's Sleep screen. The Average Blood Oxygen reading is the percentage of your blood that contains oxygen. The Breathing Regularity graph displays variations in how much oxygen is present in your blood. This is different from respiratory rate, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during sleep periods longer than three hours, but the Oura App only displays the results for the longest sleep period of the day. Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day. (https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2)
16[c][iii] refrain from the obtaining of the second information based at least in part on a determination that the	The Oura Accused Product refrains from the obtaining of the second information based at least in part on a determination that the first information satisfies a specified condition. For example, the Oura Accused Product refrains from the obtaining of the second information, for example biological information such as SpO ₂ or

	U.S. Patent No. 10,945,677
first information satisfies a specified condition.	blood oxygen levels, based at least in part on a determination that the first information satisfies a specified condition, such as the user being awake or the user's motion satisfying a condition indicative of being awake. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

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